MILITARY SPECIFICATION

TESTER, PLIABILITY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the requirements for a device which measures quantitatively the pliability of wrapping materials, i.e., the ease with which a material can be molded to conform to an object being wrapped, or the resistance of a material to a wrapping operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601 — Boxes, Wood, Cleated Plywood.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

P-101290 — Pliability Tester, General Arrangement (Side View).

P-101291 — Pliability Tester, General Arrangement (Front View).

P-101292 — Pliability Tester, General Arrangement (Top View).

P-101293 — Pliability Tester, General Arrangement (Specimen Cutter).

P-101294 — Pliability Tester, List of Parts, Specifications and Standards.

P-101295 — Pliability Tester, Details.

P-101296 — Pliability Tester, Details.

P-101297 — Pliability Tester, Details.

P-101298 — Pliability Tester, Details.

P-101299 — Pliability Tester, Details.

P-101300 — Pliability Tester, Details.

P-101301 — Pliability Tester, Details.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL ELECTRICAL CODE

(Copies may be obtained from the National Board of Fire Underwriters, 85 John Street, New York, N.Y.; 222 West Adams Street, Chicago 6, Ill.; and Merchants Exchange Building, San Francisco 4, Cal.)

FSC 6635

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3. REQUIREMENTS

3.1 Construction and assembly. The pliability tester, including the specimen cutter and calibrating weights, shall be constructed and assembled in accordance with the requirements of this specification and the applicable drawings listed in paragraph 2.1.

3.2 Materials. The materials used shall be in accordance with the requirements indicated on the applicable drawings listed in paragraph 2.1.

3.3 Wiring. All wiring and electrical equipment shall be installed in accordance with the National Electrical Code.

3.4 Workmanship. Workmanship shall be in accordance with the highest grade practices covering the manufacture of laboratory testing equipment. All parts shall be carefully and accurately machined to the tolerances indicated on the applicable drawings, to provide proper fit and clearances of the various components in making up the complete assembly. Special care shall be taken to ensure that the hemispherical dome, specimen holder, and inner ring are concentric. Construction, workmanship and parts shall be of the best quality and free from dents and scratches and from defects that would cause improper or inefficient operation. The finished pliability tester shall be free from defects which affect appearance or which may affect serviceability.

3.5 Nameplate. Each machine shall have a nameplate attached to the front panel showing the following information:

PLIABILITY TESTER

Ord. Corps Dwg. P–101290

Serial Number

Designed by Container Laboratories, Incorporated

Manufactured by

Contract

Each major component of the equipment whether produced or purchased by the contractor, shall have a distinctly stamped nameplate securely affixed, cast, or permanently attached thereto in a conspicuous place showing horsepower, capacity, speed, serial number, name of manufacturer and any other pertinent information necessary. The nameplate of a distributing agent only will not be acceptable.

3.6 Treatment. The piston dome and inner ring shall be thoroughly treated with a silicone mold release agent to facilitate cleaning.

3.7 Operating and maintenance instruction. One copy of operating and maintenance instructions shall be furnished with each tester. The operating instructions shall include the information contained in the appendix to this specification and in addition, shall give complete information of the recommended care and maintenance of the machine. It shall also include parts lists, parts numbers, type of packing and lubricants recommended, and any other information necessary for complete and satisfactory performance.

3.8 Performance. Each instrument shall, when assembled, perform in accordance with the principles of operation described in the appendix. The travel of the piston above the top of the inner ring shall be $3\frac{1}{2} \pm \frac{1}{4}$ inches.

3.9 Supplies. With each instrument, there shall be furnished 20 No. 16 Exacto cutting blades, 1 pint of silicone mold release agent, 1 pint of denatured ethyl or methyl alcohol or mineral spirits, ½ pint of the recommended lubricating oil and 2 square yards of soft cleaning cloth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the

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supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to tthe Government as specified in the contract or order.

4.2 Sampling. Each pliability tester shall be inspected for compliance with the requirements of this specification and the applicable drawings.

4.3 Testing.

4.3.1 *Visual inspection.* The instrument shall be inspected visually for compliance with the requirements of this specification and the applicable drawings.

4.3.2 *Piston travel.* The height of the travel of the piston at the top of its stroke shall be measured with a suitable height gage from the top of the inner ring.

4.3.3 *Operation.* The instrument shall be calibrated and tested for satisfactory operation as outlined in the appendix to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing, levels A, B and C. Unless otherwise specified, each pliability tester shall be packed in a cleated plywood box conforming to style A of Federal Specification PPP–B–601. The instrument shall be blocked within the container with corner or edge pads of cane-fiber cushioning material in such a manner as to provide a 2-inch clearance at sides and a ½ inch clearance at front and back. The piston dome shall be removed from the instrument and packed in a separate shipping container together with the 5 pound weight platform assembly, the 5 and 10 pound calibrating weights, and other supplies. The specimen cutter arm shall be secured to the cutter support arm. The specimen drawer, cutter door, and rear access door shall be secured with pressure-sensitive tape which will not harm the cabinet surfaces.

5.2 Marking. Each shipping container shall be marked in accordance with MIL–STD–129 and with any additional information specified in the contract or order.

6. NOTES

6.1 Intended use. This pliability tester is intended for use in measuring the effort required to wrap an item with a flexible barrier-material by a simulation of the wrapping process.

6.2 Ordering data. Contracts or orders should specify the following:

(a) Title, number and date of this specification.

(b) Quantity.

(c) Any marking required in addition to that required by MIL–STD–129.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Ordnance Corps
Navy—Bureau of Naval Weapons
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Preparing activity:

Army—Ordnance Corps

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10. SCOPE

10.1 This appendix covers the method of operation of the pliability tester in testing barrier material samples for pliability.

20. CALIBRATION

20.1 Conditioning of the instrument. Prior to calibrating, the instrument should be warm, level and running efficiently.

20.2 Calibration with twenty-pound load. The pliability tester should be started and the piston stopped at the top of its stroke. The 5 pound weight platform, the 5 pound weight and the 10 pound weight shall be placed squarely on the piston dome. Then the power switch shall be turned to ON until the gage returns to its neutral position and stops. The starter button shall be pressed and held until the piston completes its downward stroke[1] (allowing the plunger on the gage stem to come into contact with the reset assembly) and returns upward until the gage is past the window opening. This will trip the indicator needle and allow it to record an accurate reading for the weight used. This operation should be repeated at least three times to ensure smoothness of operation and reproducible readings. Record the readings.

20.3 Calibration with fifteen-pound load. The piston should be stopped at the top of the stroke, by turning the power switch off, and the 5 pound weight removed. This leaves a 15 pound load. Calibrate, with the 15 pound load, as outlined in 20.2. Record the readings.

20.4 Calibration with ten-pound load. The piston should be stopped at the top of the stroke by turning the power switch off. The 10 pound weight shall be removed and the 5 pound weight replaced. This leaves a 10 pound load. Calibrate, with the 10 pound load, as outlined in 20.2. Record the readings.

20.5 Calibration with five-pound load. The piston should be stopped at the top of the stroke, by turning the power switch off, and the 5 pound weight removed. This leaves a 5 pound load. Calibrate, with the 5 pound load, as outlined in 20.2. Record the readings.

20.6 Calibration with no load. The piston should be stopped at the top of the stroke by turning the power switch off. The 5 pound weight platform shall be removed to obtain a reading at zero weight load. Calibrate as outlined in 20.2 and record the readings.

20.7 The values of the weights, which are the true forces on the piston dome, shall be plotted against the average dial readings. The curve obtained is used to find the true forces corresponding to the dial readings that are obtained when using the instrument.

20.8 **Adjustment.** If the calibration reading at 10 pounds does not read 10 ± 0.1 pounds the following two adjustments may be made:

(a) Adjust the gage adjusting screw in the gage actuator arm to compensate for error; or

(b) Loosen dial locking screw and rotate dial to compensate for error.[2]

If after adjusting the 10 pound reading, the zero reading is off, adjustment shall be made by loosening the two socket head cap screws which hold the adjusting key at the rear of the piston sleeve and adjusting the indicator needle to the zero position. After adjusting, recalibrate and replot the calibration curve as described in 20.2 through 20.7.

30. OPERATION OF PLIABILITY TESTER.

[1] If pressure is removed from the starter button before the gage has passed the opening, the motor will stop. Then when the motor starts, the sudden jarring of the weights may cause a false reading.

[2] Procedure "b" should be followed only in the event that the locking screw loosens sufficiently to allow the dial face to get out of adjustment.

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30.1 General operation of the instrument. A sample of barrier material 10 inches in diameter is placed in the cabinet beneath the inner ring. The instrument is turned on with the toggle switch in the front panel, and specimens are tested by pushing the starter button. The hemispherical piston dome is driven upward through the ring by a geared down electric motor. As the piston dome carries the sample upward, the ring forces the barrier material around the hemispherical surface of the piston dome in a manner similar to wrapping that surface. The force that is exerted on the piston dome in accomplishing this work is a measure of the effort required to wrap the material. The magnitude of the force is measured by the deflection of a spring within the piston sleeve which is compressed by the piston dome. This deflection is transmitted through a linkage to a gage which is calibrated to indicate the force, in pounds, on the piston dome. The maximum reading remains on the gage, and the gage is automatically reset to zero at the beginning of each cycle.

30.2 General operation of specimen cutter. The specimen cutter is located in the bottom of the cabinet and must be completely removed before operating. Raise lift knob located in center to permit one or more sample sheets to be placed under pressure plate. Release lift knob and apply slight pressure to it with one hand while the other hand turns the rotating knob clockwise. Raise lift knob to remove specimens. Two or more specimens of creped barrier material may be cut at one time if the cutting blade crumples a single sheet. Lightweight paper or film may be placed under and above specimens of wax covered barrier materials to facilitate inserting and removing barrier materials from the cutter and to keep the cutter base and pressure plates free of wax. The cutter blade may be replaced by loosening the knurled screw on the end of the cutter arm.

30.3 Detailed operational procedure.

30.3.1 *Test conditions.* All tests on barrier materials shall be made under uniform atmospheric conditions of 70° to 75° F. and 48 to 52 percent relative humidity.

30.3.2 *Preparation of test specimens.* The specimens of barrier material shall be 10 inches in diameter cut from samples approximately 12 inches square which have been conditioned for 12 hours at the temperature and humidity specified in 30.3.1. The samples should be suspended in the conditioning chamber in a manner that allows the atmosphere to have access to both sides of each sample.

30.3.3 *Cleaning and maintenance of the test surfaces.* Before each test the ring and the piston dome shall be wiped clean with a soft cloth moistened with alcohol or mineral spirits to remove any traces of wax from previous tests. Although friction between the piston dome and the sample does not influence the force on the dial gage as does the friction between the ring and the specimen, it is easier to clean the piston dome after testing each specimen than it is to remove the tightly adhering wax at the completion of a number of tests. The piston dome should be treated every few months with a silicone mold release agent to facilitate cleaning before each test.

30.3.4 *Specimen testing.* The specimen of barrier material shall be placed in the cabinet drawer, positioned within the guide channels so as to be centered and flat. Five specimens of a material shall be run with the greaseproof side up and 5 with the greaseproof side down. With the power switch in the ON position, the starter button shall be pressed and held for a few seconds. The specimen drawer must be completely closed in order for the instrument to operate. The piston dome will push the barrier material upward through the ring the full length of its stroke and will return downward until the gage stops in the center of the front window

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opening. The needle of the dial gage will remain at its maximum reading. The maximum reading of the dial gage, corrected according to the calibration curve (see 20.7 and 20.8) shall be taken as the "pliability" reading of the sample. The 5 readings with the greaseproof side up shall be averaged and the 5 readings with the greaseproof side down shall be averaged. No "up" reading should deviate by more than 25 percent from the average of the 5 "up" readings. Similarly, no "down" reading should deviate by more than 25 percent from the average "down" reading. In addition, the absolute sum of the deviations, in pounds, of the 5 "up" readings and the 5 "down" readings shall not exceed 12 pounds. In taking this sum, negative deviations shall be added to the positive deviations, rather than subtracted. If the absolute sum of the deviations exceeds 12 pounds, the instrument shall be recalibrated. If an "up" or a "down" reading deviates by more than 25 percent from the average, a sixth sample shall be tested and its reading shall be used in place of the one causing the additional test. If, after averaging the sixth reading with the 4 original readings, one of the readings deviates by more than 25 percent from the average, 4 additional samples shall be tested, making a total of 10 and these 10 readings averaged. If, among the initial 5 samples in one direction, 2 or more deviate by more than 25 percent from the average of the 5, an additional 5 samples shall be tested in this direction and these 10 readings averaged. It shall be recorded that 10 samples rather than the normal 5 were required. The average of the "up" readings and the average of the "down" readings shall be averaged together and the result recorded as the "pliability" value for the material. This procedure will ensure that the probable error in the true pliability value will not exceed $\frac{1}{3}$ of a pound.

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